

# Project Design Phase-II: Solution Requirements (Functional & Non-functional)

**Date:** 30 September 2026  
**Team ID:** SWTID-2026-1128  
**Project Name:** Auto Ticket Classification using Flow Designer  
**Maximum Marks:** 4 Marks

---

## Functional Requirements

Following are the functional requirements of the proposed Auto Ticket Classification using Flow Designer solution:

| FR No. | Functional Requirement (Epic)       | Sub Requirement (Story / Sub-Task)                                                                                                                                                                               |
|--------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-1   | User Authentication & Portal Access | Login via Single Sign-On (SSO / University Credentials)<br>Role-based access control for Students, Faculty, Agents, and Admins                                                                                   |
| FR-2   | Ticket Submission                   | Submit IT support ticket via Service Portal<br>Input fields for Short Description, Detailed Description, and Attachment upload                                                                                   |
| FR-3   | Automated Flow Designer Triage      | Trigger Flow Designer workflow on new incident record creation<br>Keyword extraction and pattern matching against Short Description / Description<br>Automated assignment of Category, Subcategory, and Priority |

|      |                              |                                                                                                                                                                                          |
|------|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-4 | Smart Routing & Notification | Automatic routing of classified tickets to respective Assignment Groups (e.g., Network, Hardware, Software)<br>Instant email notification sent to the end user and assigned support team |
| FR-5 | Queue & Exception Management | Support executive dashboard for viewing pre-classified queue<br>Manual override and re-classification capability for edge-case tickets with audit trail                                  |

## Non-Functional Requirements

Following are the non-functional requirements of the proposed solution.

| FR No. | Non-Functional Requirement | Description                                                                                                                                                                                   |
|--------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NFR-1  | Usability                  | The ServiceNow Service Portal and dashboard must provide an intuitive, responsive, and accessible user interface for students, faculty, and support agents across desktop and mobile devices. |
| NFR-2  | Security                   | Ensure strict role-based access control (RBAC), data encryption in transit (HTTPS/TLS) and at rest, and secure API credentials adhering to university IT security standards.                  |
| NFR-3  | Reliability                | The Flow Designer classification workflows must execute reliably without data loss or pipeline failures, with                                                                                 |

|       |              |                                                                                                                                                                               |
|-------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       |              | robust error-handling and fallback default routing for ambiguous descriptions.                                                                                                |
| NFR-4 | Performance  | Flow Designer automated ticket classification and routing must complete within 3 seconds of ticket submission to ensure near real-time queue assignment.                      |
| NFR-5 | Availability | The system must maintain 99.9% uptime during operational academic hours, leveraging ServiceNow cloud infrastructure redundancy.                                               |
| NFR-6 | Scalability  | The architecture must seamlessly handle peak ticket volumes during semester beginnings and exam periods without degradation in classification speed or system responsiveness. |